

Circular Structures of Puffer Fish: A New Metaheuristic Optimization Algorithm

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Abstract— In this paper, we presented a new metaheuristic optimization algorithm which we named as circular structures of puffer fish. This algorithm has been inspired by the behaviors of male puffer fish for charming the female ones. The special type of male puffer fishes, *Torquigener albomaculosus*, has a special ability to build a unique circular structure on the seabed. The building process of a circular structure consists of several regular sub-dimensions and as the result of this process, a circular structure is constructed. The water flow rate at the center of this structure is reduced. Additionally, these structures have some numerical properties such as the ratio of inner and outer circular structures diameter or the number of peaks on the structure. The results of the proposed algorithm are compared with genetic algorithm through various performance criteria. The most popular performance analyzing functions were used in the optimization of the algorithm performance comparison process.

Keywords—component; optimization problem, metaheuristic algorithm, bio-inspired algorithms, genetic algorithm, adaptive search

I. INTRODUCTION

Metaheuristic or bio-inspired optimization algorithms are inspired by some patterns of natural life animals' mating behaviors, insects' food collection [1,2]. In this paper, a novel metaheuristic optimization algorithm was determined and it was named as 'circular structures of puffer fish'. The metaheuristic algorithms have several advantages according to numerical optimization methods. The most important advantage is that they are adaptable to the different types of problems. The metaheuristic algorithms have the ability to adapt solution space for determination of optimal solution.

Numerical optimization algorithms can provide satisfying results on convex optimization problems [3]. However, in engineering, most of the optimization problems do not have a convex characteristic, furthermore, they have several local minimums [4]. For such reasons, researchers have sought to find different algorithms that will give satisfactory results for different types of problem, such as for pattern recognition, signal processing [5] and power electronics.

Firstly, the problem-solving skills of creatures have been examined in nature [6]. In 1960, Rechenberg suggested an algorithm for aircraft wing sections via actual values.

Afterward, several metaheuristic optimization algorithms were suggested based on biological evolution. In parallel to

advances in the computational ability of the computers, the popularity of metaheuristic methods has increased day by day.

According to general acceptance, the modern history of metaheuristic optimization algorithms started with genetic algorithm. This method is inspired by the process of transferring the most successful genes to the next generation [7,8]. John Holland studied the genetic algorithm in his study: *Adaptation in Natural and Artificial Systems* in 1975.

In this study, the process of adaption in nature and biological evolution were examined in the light of mathematical language. After this study, the developed methods for metaheuristic algorithms are of this origin. Although many methods have been developed after this process, some of them became quite popular and have been adopted in many areas of use. These methods have been carried out modeling of the process for finding food in nature. Firstly, ant colony optimization algorithm was performed in 1992 [9].

The location determination of food was modeled in the method. The main contribution of this optimization algorithm is the parameter of pheromone. This chemical messenger parameter has provided fore-information about the location of food for ants and this parameter was integrated into mathematical modeling for correct and rapid determination of global extrema of the function. Another important metaheuristic optimization algorithm is particle swarm intelligence (PSO) method [10,11].

The method inspired by the behaviors of fish or bird in nature. This algorithm has several similarities with genetic and ant colony algorithms. But, it is simpler than the genetic and the ant colony algorithms. Because there are no complex parameters such as crossover/mutation or pheromone in this method. Additionally, the real random numbers are used in PSO. After that, the swarm intelligence based metaheuristic algorithms have become very popular in the area of metaheuristic optimization algorithms.

Today, most of the metaheuristic optimization algorithms are inspired by swarm intelligence and these are usually the most successful ones in the literature. Another popular metaheuristic optimization algorithm is artificial bee colony (ABC) optimization [12]. This method inspired by nectar search process of bee. The bees have a division of labor in beehive such as observer and employer bees.

When the bees find suitable nectar, they give information to other bees via dances. In mathematical modeling, this reaction can be implemented via coefficients and equations. Unfortunately, most of the metaheuristic optimization algorithms need random process for avoiding the stuck local minimums at models. The solution for this problem can only be provided via stochastic process [13]. In this case, the performance of metaheuristic optimization algorithms depends on the success of the random number generators. However, the achievement rate of algorithms is highly related to balanced use of the stochastic and deterministic processes [14]. Therefore, most of the successful algorithms in global optimization have to include random and non-random processes at problem-solving stages. How to determine the distribution of these two processes with correct mathematical modeling or coefficients is important. This article is inspired by the construction process of circular objects built by the puffer fish.

II. PUFFER FISH

In this paper, special type of puffer fish, *Torquigener albomaculosus* (Figure 1) has been studied. [15]. A very special ability of this fish species has been discovered recently. The male of this fish species has ability to build unique and charming circular structures on the seabed and these structures were mysterious for many years.



Figure 1. *Torquigener albomaculosus*

In the recent studies, the process and realization of these structures could be examined in detail. The one of the circular structure which was built by male puffer fish is shown in Figure 2 [16].



Figure 2. Mysterious circular structure

These structures had been the mystery of the seaside for many years. In recent years, the studies have revealed how and by which organism these structures were constructed. The findings were very interesting for researchers. These mysterious structures are constructed by male puffer fishes to impress female fishes and diameter of structure can be about 2 meters [16,17]. Additionally, all the circular structures have a unique pattern within itself. The circular structures have several rules at the building stage. Also, it is built on a specific set of rules. The construction process can be divided into three stages that are early, middle and final stages. In the early stage of construction, the male puffer fish moves in the center of circular structure and creates a pattern for peaks. In the middle stage, the peaks and valleys become more obvious and the structure becomes more obvious and at the final stage, the puffer fish creates a random pattern at the center. As a result, the flow rate of water decreases 24.2% at the center zone [16]. The type of sands are different for different zones, large sand pieces are used at the outer zones and small size sands are used at the center zones. With this information, the circular structures which are created by male puffer fish might be adapted to solve a minimization problem as a metaheuristic optimization algorithm. In this study the proposed algorithm and the test results confirmed this prediction.

III. CIRCULAR STRUCTURES OF PUFFER FISH

In this section, the implementation of new metaheuristic optimization algorithm which is named as circular structures of puffer fish (CSOPF) is explained. The algorithm has steps similar to the real construction of circular structures by male puffer fish. Initially, the users have to determine some parameters for circular structures. These are the following: number of peaks, initial points, radius of inner and outer zones and number of iteration like other metaheuristic optimization algorithm. In the center zone, random centripetal patterns are created. Additionally, new parameter has to be added to increase the success rate of algorithm and it has been called as radius reduction ratio of this optimization algorithm. The illustration of circular structures is shown in Figure 3.

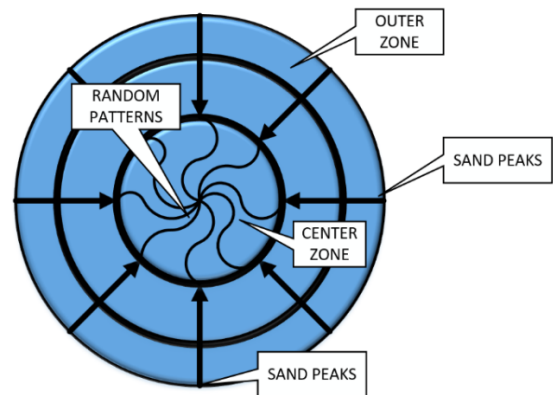


Figure 3. Circular object

The diameter ratio between the center and outer zones and the number of peaks were adopted from real structures of puffer fish. The convergence of the method that is provided via the lowest points calculated at each iteration is determined as the new starting points. In this way, a global minimum point of function can be converged successfully. The parameters of the algorithm are defined as follows: the radius of outer zone (r_{out}), center zone (r_{cen}), number of peaks (n_{peak}) and radius reduction ratio (d_{red}). The implementation of proposed algorithm is similar to the other metaheuristic algorithms. At the beginning, the users have to define some initial parameters.

The size of sands is also modeled in proposed algorithm. Large sands (s_{large}) are modelled via big step size at outer zone and small pieces of sands (s_{small}) are modelled at small step size in center zone. c_{max} is maximum number of circular structure. The parameter of r_c is defined as specific value of radius according to the number of circular structures and it is updated according to Equation 1.

$$r_{c+1} = d_{red} \times r_c \quad (1)$$

The pseudo code of proposed algorithm is shown below.

Circular structures of puffer fish algorithm

Begin

The objective function $f(x)$, $x = (x_1, x_2, \dots, x_d)^T$
 Define Initial Parameters (r_{out} , r_{cen} , n_{peak} , d_{red})
 Initialize location vector (x_i , y_i)
 Maximum number of circular structures (s_{max})
 Error criteria (ϵ)

While ($c < c_{max}$) or satisfied error criteria

Evaluate its quality/fitness
 Update radius according to Equation 1.

If ($r_s < r_{cen}$)

Increase the solution resolution (step size) and
 create random pattern in center zone

End

Create patterns oriented to the center of the
 circular structure via sand peaks
 Use optimal locations for new initial locations
 $c = c + 1$

End while

IV. NUMERICAL SIMULATION RESULTS

The performance of the proposed algorithm was compared with genetic algorithm (GA). In this study, 14 of common optimization test functions were used to test the new algorithm. In the test process, two different performance criteria were examined, these are mean computation time and mean error between real coordinates and results of algorithms. Besides, some of the coefficients of proposed algorithm were selected in the light of the real circular structures at the seabed. For example, the ratio between outer and center zone was selected $r_{out}/r_{center} = 1.5$. Additionally, the

number of sand peaks and step size ratio were defined in a similar way. The initial parameter for circular structures of puffer fish algorithm (CSOPF) is shown in Table 1.

TABLE I. PARAMETERS FOR PROPOSED ALGORITHM

n_{peak}	24
c_{max}	50
r_{out}/r_{center}	1.5
s_{large}/s_{small}	5
d_{red}	0.7

The GA parameters were selected as follows; probability of crossover and mutation ratio are $p_{crossover} = 0.9$ and $p_{mutation} = 0.1$, respectively. The maximum number of iteration is selected as 50 and the initial population is equal to 24, to be similar to the proposed method. One of the test function which is called as Schwefel is shown in Figure 4. In this function, there are several local minimum points and the value of global minimum is very close to the other local minimums. The mathematical definition of Schwefel function for 2D is shown in Equation 2 [18].

$$f(x, y) = -x \sin \sqrt{|x|} - y \sin \sqrt{|y|} \quad (2)$$

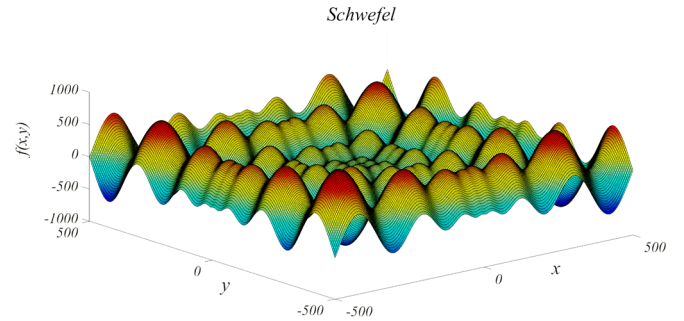


Figure 4. Test function Schwefel

The performance comparison result of Schwefel function is shown in Table 2.

TABLE II. SCHEWFFEL TEST RESULTS

GA	CSOPF	Real Values
$\min_{GA}(f(x, y))$ = -837.3587	$\min_{CSOPF}(f(x, y))$ = -837.8657	$\min_{Real}(f(x, y))$ = -837.9657
$x_{min} = 419.553$ $y_{min} = 422.647$	$x_{min} = 420.673$ $y_{min} = 421.137$	$x_{min} = 420.968$ $y_{min} = 420.968$
Mean Computation Time (second) 1.45482	Mean Computation Time (second) 1.079412	

As shown in the table, the proposed algorithm has better performance results compared to GA. The change of cost function (Schwefel) values versus number of circular structures is shown in Figure 5.

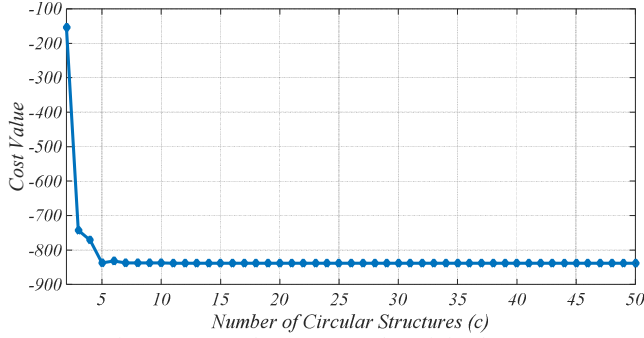


Figure 5. Cost value versus number of circular structures

As shown in the figure, the real minimum value of function can be easily obtained with a small number of circular structures. The change of circular structures according to iteration number is shown in Figure 6. As shown in the figure below, the radius of the circular structure decreases with iteration and the random pattern starts below a certain value at center zones. Additionally, the step size of calculating point is increasing in the center zone. This increase is representing the size of sands at circular structures.

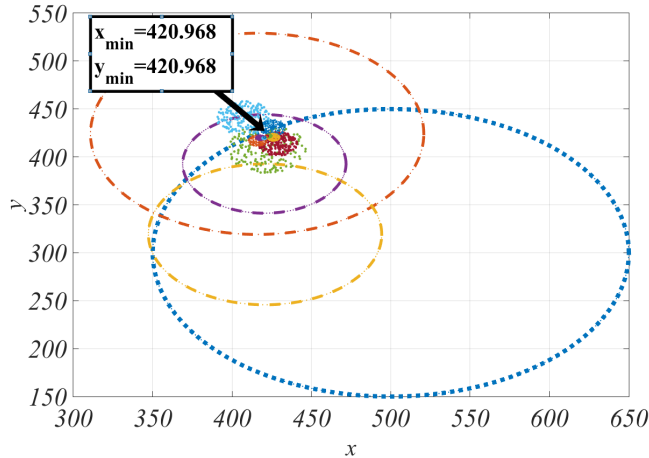


Figure 6. Change of circular structures

The one of main contribution of the algorithm is that most of the coefficients are inspired by real circular structures of puffer fish in nature. The determination of parameters about metaheuristic algorithms is one of the most difficult processes. These difficulties have been eliminated by natural inspiration in the proposed method. Moreover, the proposed method reduces complexity compared to other metaheuristic algorithms and it is suitable for parallel computation process.

The benchmark results of the proposed algorithm with common functions at literature are shown in Table 3 [19].

TABLE III. RESULTS OF BENCHMARK TEST FUNCTIONS

Functions	$\min_{\text{Real}}(f(x, y))$	$\min_{\text{GA}}(f(x, y))$	$\min_{\text{CSOPF}}(f(x, y))$
Ackley	0	4.4421×10^{-7}	1.3793×10^{-9}
Booth	0	4.2295×10^{-9}	1.4978×10^{-11}
Bukin4	0	∞	2.8847×10^{-14}
Carromtable	24.1568	22.683271349	23.7193876783
Crownedcross.	0.0001	0.0488685445	0.00516543546
Greiwank	0	5.0038×10^{-13}	1.3617×10^{-16}
Himmelblau	0	1.5698×10^{-13}	4.1743×10^{-17}
Leon	0	1.233×10^{-11}	6.178×10^{-14}
Levi13	0	1.9841×10^{-15}	3.1793×10^{-24}
Mccormick	-1.91946322	-1.9105077594	-1.9134985437
Modschaffer1	0	0.00135664519	0.000001389764
Modschaffer2	0	0.00314363513	0.000007156756
Modschaffer3	0.00156685	0.00173643221	0.001593615677

V. CONCLUSIONS AND FUTURE WORKS

In this paper, a novel metaheuristic algorithm which is called as circular structures of puffer fish was performed. This algorithm is inspired by the behaviors of male puffer fish. The name of fish is *Torquigener albomaculosus* and it has a great architectural ability. This ability inspired the new algorithm.

The novel algorithm has two main stage, in the first stage circular structures with peak points are created according to real structures and in the second stage, random patterns are created as a special zone which is called center zone. Additionally, the coefficient of algorithm is also inspired by real structures constructed by male puffer fish in nature. The sand sizes at the center and outer zone are modeled with step size in the algorithm. The performance of the proposed algorithm is compared with genetic algorithm via common benchmark functions in the literature. The proposed algorithm has better performance and lower computation time. The computation time decreasing about 22% according to the genetic algorithm.

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